

Day 22 had real progress! I got the Windows Server running.

Today's focus is static IP and Active Directory Domain Services Prep

So today's mission is moving from the Windows Server just existing to Windows Server being prepared to become a Domain Controller.

Main goals:

1. Verify server networking
2. Configure static IP
3. Understand why DNS matters
4. Install AD DS Role
5. Begin Domain Controller promotion (if time allows)

### **Pre-Flight Check**

Run hostname in Powershell. Run ipconfig /all.

Hostname: SRV-DC01  
IPv4 Address: 192.168.1.236  
Subnet Mask: 255.255.255.0  
Default Gateway: 192.168.1.1  
DNS Servers: 192.168.1.1  
DHCP Enabled: Yes

After run: Get-NetAdapter. Write down the adapter name exactly.

Adapter Name: Ethernet0

Run: **Get-NetIPAddress -AddressFamily IPv4**. Identify main IPv4 address attached to active network adapter.

### **Static IP Concept Before Touching Settings**

Domain controllers: they need to have a static IP because computers on the domain need to find it. This could be for: DNS, Kerberos, LDAP, Authentication, Group Policy, Domain login

If the DC's IP address isn't static (keeps changing), clients may not know where to send login requests.

Client PC -> "Where is my DC?" -> DNS point to SRV-DC01 -> SRV-DC01 must stay at a predictable IP address....

Why should a Domain Controller use a static IP address?

Other clients or servers need to find it reliably every time.

- AD login
- DNS

## **Configure Static IP**

GUI first...First lab with Windows Server.

Server Manager -> Local Server -> Click On Network Adapter Identified (Ethernet 0) -> Properties -> Internet Protocol Version 4 -> Properties -> Set IP Address and DNS

DHCP should now say it is not enabled, you can test this using `ipconfig /all`. Ping 8.8.8.8 for connectivity test.

Ping google.com, shouldn't work

## **Install AD DS Role**

Server Manager -> Manage -> Add Roles and Features

Role-based or feature-based

Select Server

## **Understand Promotion Before Doing It**

What does it mean to promote a Windows Server to a Domain Controller?

Turning a normal Windows Server into a server that controls and manages an Active Directory domain.

Promotion means:

Normal Windows Server -> Domain Controller -> Hosts An Active Directory Database -> Authenticate Domain Users -> Provides Domain Services

What is a new forest in AD?

Creating a completely new, independent Active Directory environment from scratch.

Domain Name in AD: human readable name used to identify network, website, or AD domain

Mine: kevinlab.local

Forest: New Forest

Domain name: kevinlab.local

Domain Controller Name: SRV-DC01

## **Start Domain Controller Promotion**

Click on the promotion button within, and follow the Wizard.